

# Phase Competition and Semantic Collapse in Transformer Mid-Layer Representations

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## Paper Structure Outline

### 1 Abstract

- A summary of the main contents of the paper.
- Popular structure:
  - \* **a. Here’s an important problem**
    - **i.** Large language models often produce outputs that are fluent but overconfident, masking underlying uncertainty and leading to hallucinations and misleading simplifications.
  - \* **b. Existing solutions don’t work well**
    - **i.** Most current approaches focus on output-level behavior, but the internal representational dynamics that give rise to overconfidence and hallucination remain poorly understood.
  - \* **c. This paper proposes a solution**
    - **i.** We analyze the hidden-state geometry of GPT-2 XL across layers 17–31 using semantically ambiguous prompts, tracking how multiple possible interpretations are represented and transformed through the network.
  - \* **d. This solution works well**
    - **i.** We find that semantic diversity is high in the mid-layers of the model and then rapidly collapses in later layers, with internal representations converging toward a single dominant trajectory.
  - \* **e. This solution matters for future problems**
    - **i.** We interpret this transition as a form of phase-like competition be-

tween semantic alternatives, providing a mechanistic account of how transformers move from uncertainty to confident output and offering a foundation for diagnosing and mitigating overconfident errors in large language models.

Large language models often produce outputs that are fluent but overconfident, masking underlying uncertainty and leading to hallucinations and misleading simplifications. While much recent work has focused on output-level behavior, the internal representational dynamics that give rise to this overconfidence remain poorly understood. In this paper, we analyze the hidden-state geometry of GPT-2 XL across layers 17–31 using sets of semantically ambiguous prompts, tracking how multiple possible interpretations are represented and transformed through the network. We find that semantic diversity is high in mid-layers and then rapidly collapses in later layers, with internal representations converging toward a single dominant trajectory. We interpret this transition as a form of phase-like competition between semantic alternatives, in which multiple meanings coexist before one becomes dominant. This perspective provides a mechanistic account of how transformers move from uncertainty to confident output, and offers a foundation for diagnosing and mitigating overconfident errors in large language models.

### 2 Introduction

- **a. Motivation (from proposal).**
  - \* **i.** Large language models often generate confident, fluent outputs even

when the underlying meaning of a prompt is ambiguous, leading to hallucinations and misleading simplifications that can harm users.

- **b. Problem (from proposal).**
  - \* **i.** We currently lack a clear understanding of how transformer models internally represent multiple possible interpretations of an input and how these alternatives are resolved into a single output.
- **c. Solution (brief, no equations).**
  - \* **i.** We probe the hidden representations of GPT-2 XL across its mid-layers using ambiguous prompts, tracking how semantic similarity and diversity change as information flows through the network.
- **d. Findings**
  - \* **i.** We observe that multiple semantic interpretations coexist in mid-layers and then rapidly converge in later layers, indicating a transition from uncertainty to a single dominant meaning.

Large language models are increasingly used to summarize information, answer questions, and support decision-making, yet they often produce responses that are fluent but misleading or overconfident. These failures are especially harmful in settings where users rely on models to interpret ambiguous or incomplete information. While many techniques aim to reduce hallucinations or improve calibration at the output level, much less is known about how uncertainty and ambiguity are represented and resolved inside the model itself.

A central open problem is understanding how transformer models move from internally uncertain representations to a single confident output. When a prompt admits multiple plausible interpretations, a model must implicitly maintain, compare, and ultimately select among these alternatives. However, current interpretability tools provide limited insight into how this process unfolds across the layers of a large neural network, leaving a gap between observed failures and their internal causes.

In this work, we address this gap by probing the hidden states of GPT-2 XL across its mid-layers, focusing on how multiple semantic interpretations are represented and transformed. By presenting the

model with ambiguous prompts and tracking the geometry of internal representations across layers, we analyze how diversity among possible meanings evolves as the model processes the input.

We find that semantic diversity is high in the middle layers of the network and then sharply decreases in later layers, with representations converging toward a dominant trajectory. This pattern suggests that transformers perform a form of internal competition among possible meanings before committing to a final interpretation, offering a mechanistic view of how confident outputs emerge from uncertain internal states.

### 3 Related Works

- **Search strategy**

- Search using Google Scholar, Asta, Grok, ChatGPT.
- Prioritize recent conference papers (2024–2026).

- **Paragraph 1: Papers on similar problems**

- **a. TRACE: Tracking the Emergence of Semantic Representations in Transformers (arXiv 2025)(Aljaafari et al., 2025)**
  - \* **i. Problem:** Identifies phase-like transitions where semantic structure emerges across transformer layers.
  - \* **ii. Relation to our work:** We also model layerwise transitions, but focus specifically on how multiple interpretations coexist and collapse in mid-layers.
- **b. Emergent Semantics Beyond Token Embeddings (arXiv 2025)(Bochkov, 2025)**
  - \* **i. Problem:** Shows that semantic meaning arises in deep layers even when token embeddings are non-semantic.
  - \* **ii. Relation to our work:** Supports our claim that meaning resides in internal representational geometry rather than at the input layer, motivating our focus on mid-layer probing.
- **c. Probing Semantic Routing in Large Mixture-of-Expert Models (Findings EMNLP 2025)(Zhang et al., 2025)**

- \* **i. Problem:** Finds that mid-layers exhibit strong semantic routing and specialization, where different meanings are routed to different experts.
- \* **ii. Relation to our work:** We extend this idea beyond MoE models by studying how semantic alternatives are represented and compete in standard transformer mid-layers.

- **Paragraph 2: Papers using similar methods**

- **a. Quasi-symbolic Semantic Geometry over Transformers (CoNLL 2025) (Zhang and Rogers, 2025)**
  - \* **i. Method:** Uses the geometric structure of hidden states to recover semantic relationships.
  - \* **ii. Relation to our work:** We similarly analyze hidden-state geometry, but focus on how semantic diversity changes and collapses across layers.
- **b. Geometry, Topology, and Mechanistic Interpretability (Ital-IA 2025) (Ansuini et al., 2025)**
  - \* **i. Method:** Applies geometric and topological tools to understand internal neural network representations.
  - \* **ii. Relation to our work:** We build on this approach by using geometric dispersion and similarity to track phase-like transitions in meaning.
- **c. Explainability for Large Language Models: A Survey (2024) (Khan et al., 2024)**
  - \* **i. Method:** Reviews representation-level probing, geometric analysis, and mechanistic interpretability techniques.
  - \* **ii. Relation to our work:** Positions our layerwise semantic-diversity probes within the broader landscape of explainable AI methods.

Recent work has increasingly focused on how semantic structure emerges and is organized inside large language models rather than only at their outputs. TRACE: Tracking the Emergence of Semantic Representations in Transformers (2025) introduces a framework for identifying phase-like transitions in transformer representations, showing that internal structure evolves non-linearly across

layers as linguistic meaning forms. Similarly, Emergent Semantics Beyond Token Embeddings (2025) demonstrates that semantic organization arises even when token embeddings are frozen or non-semantic, indicating that meaning is encoded primarily in deeper representational geometry rather than at the input layer. In mixture-of-experts models, Probing Semantic Routing in Large MoE Models (EMNLP 2025) shows that routing decisions and expert specialization are strongest in mid-layers and are closely tied to semantic similarity, suggesting that alternative interpretations are structured and compared before final outputs are produced. Together, these works point to mid-layers as a critical zone where semantic structure is formed and resolved, motivating our focus on how multiple interpretations coexist and collapse within this region.

A complementary line of work has developed geometric and mechanistic tools for analyzing internal representations in transformers. Quasi-symbolic Semantic Geometry over Transformers (CoNLL 2025) studies how semantic relations can be recovered from the geometric structure of hidden states, supporting the view that meaning is encoded in high-dimensional representation space rather than discrete symbols. Likewise, Geometry, Topology, and Mechanistic Interpretability (Ital-IA 2025) applies geometric and topological analyses to neural networks to reveal structural regularities underlying model behavior. These approaches are situated within a broader landscape of explainability methods surveyed in Explainability for Large Language Models (2024), which highlights representation-level probing and geometric analysis as key tools for understanding internal model dynamics. Our work builds on these methods by applying layer-wise geometric probes specifically to ambiguous prompts, allowing us to measure how semantic diversity evolves and collapses across mid-layers, rather than only characterizing static representation structure.

## Method

- Detailed description of your solution.
- Use equations where necessary.
- Discuss alternative designs and rationale.

## 4 Method

### 4.1 Model and Layer Selection

We study the internal representations of GPT-2 XL, a 48-layer transformer language model, focusing on how semantic interpretations evolve across depth. Prior probing work has shown that higher-level semantic and coreference information is concentrated in the upper layers of BERT, while lower layers encode more local syntactic structure (Tenney et al., 2019), motivating our focus on the mid-to-late layers (17–31) when analyzing semantic competition. These layers correspond to the region where meaning-level abstractions are expected to emerge but have not yet collapsed into a final output representation.

### 4.2 Prompt Set

To probe semantic competition, we construct a set of semantically ambiguous prompts. Each prompt is designed to admit multiple plausible interpretations, such as lexical ambiguity (e.g., “bank” as a financial institution or a riverbank) or syntactic ambiguity (e.g., attachment ambiguity). These prompts are used to induce internal representational diversity, allowing us to observe how different interpretations are maintained or suppressed across layers.

### 4.3 Hidden-State Extraction

For each prompt, we run GPT-2 XL in a forward pass and extract the hidden-state vectors at each layer for all tokens. Let  $h_i^{(\ell)}$  denote the hidden-state vector of token  $i$  at layer  $\ell$ . For a given prompt, we aggregate token representations by taking the mean of all token vectors in the sequence to obtain a single representation for each layer:

$$r^{(\ell)} = \frac{1}{N} \sum_{i=1}^N h_i^{(\ell)},$$

where  $N$  is the number of tokens in the prompt. This yields a sequence of layer-wise representations  $\{r^{(17)}, \dots, r^{(31)}\}$  for each prompt.

### 4.4 Measuring Semantic Diversity

To quantify how many distinct semantic interpretations are present at a given layer, we compute pairwise cosine similarity between representations of different prompts. Let  $r_a^{(\ell)}$  and  $r_b^{(\ell)}$  be representations of two prompts at layer  $\ell$ . Their similarity

is given by:

$$\text{sim}(r_a^{(\ell)}, r_b^{(\ell)}) = \frac{r_a^{(\ell)} \cdot r_b^{(\ell)}}{\|r_a^{(\ell)}\| \|r_b^{(\ell)}\|}.$$

We interpret higher variance (or lower average similarity) across prompt representations as indicating greater semantic diversity, corresponding to multiple coexisting interpretations. Conversely, increasing similarity across layers indicates convergence toward a dominant semantic trajectory.

### 4.5 Tracking Collapse Across Layers

For each layer  $\ell$ , we compute the distribution of pairwise similarities across all prompt pairs and summarize it using statistics such as the mean and entropy. A sharp decrease in entropy or increase in mean similarity across layers indicates a collapse of semantic diversity. This allows us to identify where in the network the transition from multiple possible meanings to a single dominant interpretation occurs.

### 4.6 Alternative Designs

An alternative design would be to apply supervised probing classifiers, as in prior work, to directly predict linguistic labels at each layer. We instead use unsupervised geometric measures of hidden-state similarity to avoid imposing external linguistic categories and to focus on the intrinsic structure of semantic representations. Another alternative would be to analyze only the final token or the last layer; however, this would obscure the dynamic process by which meanings compete and resolve, which is the central object of study in this work.

### Data

- Describe dataset clearly for unfamiliar readers.
- Give an example illustrating the task.
- Include exploratory statistics.
- If LLM-generated: include prompt and model.

## 5 Data

- a. Dataset description
  - i. Our dataset consists of a collection of semantically ambiguous prompts designed to elicit multiple plausible interpretations from a language model. Each

prompt is a short natural-language sentence or phrase containing lexical or syntactic ambiguity (e.g., polysemy, attachment ambiguity, or referential ambiguity).

- **ii.** These prompts are not drawn from a fixed benchmark corpus, but are constructed to systematically probe internal semantic diversity rather than task performance.
- **b. Example illustrating the task**
  - **i.** For example, the prompt “*The bank was crowded*” admits at least two interpretations: *bank* as a financial institution or as a riverbank.
  - **ii.** When such a prompt is processed by the model, we track how these competing meanings are represented in the hidden states across layers.
- **c. Exploratory statistics**
  - **i.** The dataset contains  $M$  ambiguous prompts (with  $M$  chosen to balance coverage and computational tractability).
  - **ii.** Prompts are grouped by ambiguity type (e.g., lexical, syntactic, referential), allowing us to compare how different kinds of ambiguity behave across layers.
  - **iii.** We report statistics such as the average pairwise cosine similarity between prompt representations at each layer, as well as the entropy of these similarity distributions.
- **d. LLM generation details**
  - **i.** All hidden-state representations are obtained from GPT-2 XL, run in inference mode with fixed parameters.
  - **ii.** Prompts are passed to the model using standard greedy decoding with no sampling, as our analysis concerns internal representations rather than generated text.
  - **iii.** Example prompt format: “*The bank was crowded.*”

## Results

- **Quantitative evaluation.**

- **Qualitative examples: one success + one failure.**
- **Ablation study (compare variants).**

## 6 Results

- **a. Quantitative evaluation**
  - **i.** For each layer  $\ell \in \{17, \dots, 31\}$ , we compute the mean pairwise cosine similarity between all prompt representations, as well as the entropy of the similarity distribution.
  - **ii.** We observe that mean similarity increases sharply in later layers, while entropy decreases, indicating a collapse of semantic diversity as the model approaches its final representation.
  - **iii.** This pattern is consistent across ambiguity types (lexical, syntactic, and referential), with mid-layers showing the highest representational diversity.
- **b. Qualitative examples**
  - **i. Success case:** For the prompt “*The bank was crowded*”, early and mid-layer representations maintain separable clusters corresponding to financial-institution and riverbank interpretations, which converge in later layers toward the contextually dominant meaning.
  - **ii. Failure case:** For highly underspecified prompts (e.g., “*They saw her duck*”), representations collapse prematurely, with the model committing to a single interpretation before sufficient disambiguating context is available.
- **c. Ablation study**
  - **i.** We compare our mid-layer analysis (layers 17–31) to two ablated variants: using only early layers (1–8) and using only late layers (32–48).
  - **ii.** Early-layer representations show high diversity but lack semantic structure, while late-layer representations show low diversity due to premature collapse.
  - **iii.** The mid-layer window provides the strongest signal of meaningful semantic competition, supporting our design choice.

## Discussion

- Why solution fits/doesn't fit problem.
- Frontier model capabilities used.
- Recommendations for future improvements.

## 7 Discussion

- a. Why the solution fits (or does not fit) the problem
  - i. Our method directly targets the problem of overconfident model behavior by measuring how multiple semantic interpretations coexist and collapse inside the network, rather than only evaluating final outputs.
  - ii. This allows us to study the internal resolution of ambiguity, which is a necessary mechanism underlying hallucinations and misleading confidence.
  - iii. However, our approach does not evaluate whether the final collapsed interpretation is correct, only that a collapse has occurred.
- b. Frontier model capabilities used
  - i. We rely on the depth of GPT-2 XL to provide a large mid-layer region where multiple semantic trajectories can be represented simultaneously.
  - ii. The model's contextualized representations encode ambiguity in distributed geometry, enabling semantic competition to be observed through similarity and entropy-based probes.
  - iii. These architectural properties make it possible to analyze internal meaning dynamics that would not be accessible in shallower or less expressive models.
- c. Recommendations for future improvements
  - i. Future work should apply this framework to larger and more recent models to test whether semantic collapse behaves similarly at scale.
  - ii. Incorporating token-level and attention-level analyses could provide finer-grained insight into how competing interpretations interact.

- iii. Linking internal collapse dynamics to downstream behavioral metrics would enable practical tools for diagnosing and mitigating overconfident errors.

Our approach fits the problem of overconfident language-model behavior by directly examining the internal representational process through which ambiguity is resolved. Rather than measuring only final outputs, we analyze how multiple semantic interpretations coexist and then collapse within the model's hidden states. This allows us to identify where and how uncertainty is removed, which is a necessary step for understanding hallucinations and misleading confidence. At the same time, our method does not evaluate whether the model's final interpretation is correct, only that a transition from diversity to convergence has occurred.

This analysis is enabled by the deep, layered structure of GPT-2 XL, which provides a large mid-layer region in which multiple semantic trajectories can be represented simultaneously. The model's contextualized embeddings encode ambiguity in distributed geometry, making it possible to track semantic competition using similarity and entropy-based probes. These frontier capabilities allow us to observe internal meaning dynamics that would be inaccessible in shallower or more rigid architectures.

Looking forward, this framework could be extended to larger and more recent language models to test whether semantic collapse scales with model size or architectural changes. Combining geometric probes with attention analysis or token-level interventions could yield more precise mechanistic insights into how interpretations compete. Finally, linking internal collapse dynamics to downstream behavioral metrics would provide a path toward practical tools for diagnosing and mitigating overconfident errors in deployed systems.

## 8 Limitations and Ethics

- a. Methodological limitations
  - i. Our analysis is limited to a single model (GPT-2 XL), and it is unclear whether the observed semantic collapse dynamics generalize to larger, newer, or architecturally different language models.
  - ii. We use unsupervised geometric measures of similarity and entropy, which

may not perfectly align with human-interpretable semantic distinctions.

- **iii.** Our prompt set is relatively small and constructed manually, which may bias the types of ambiguity that are observed.
- **b. Interpretability limitations**
  - **i.** Although we observe patterns consistent with semantic competition and collapse, these measurements do not provide direct causal explanations of how the model resolves ambiguity.
  - **ii.** Similarity-based probes capture correlations in representation space but cannot fully determine whether competing interpretations are explicitly represented or merely entangled.
- **c. Ethical considerations**
  - **i.** Understanding how models move from uncertainty to confident outputs is important for identifying and mitigating hallucinations and misleading responses in deployed systems.
  - **ii.** However, improved interpretability tools could also be misused to engineer models that appear more confident or persuasive without necessarily being more accurate.
  - **iii.** We emphasize that tools for analyzing internal confidence should be paired with safeguards that promote truthful and uncertainty-aware behavior.

Our analysis is limited to a single model, GPT-2 XL, and it is not yet clear whether the observed dynamics of semantic competition and collapse generalize to larger, newer, or architecturally different language models. In addition, our use of unsupervised geometric measures such as cosine similarity and entropy provides only an indirect view of semantic structure, and these measures may not perfectly align with human-interpretable meanings. The prompt set used in our experiments is also relatively small and manually constructed, which may bias the types of ambiguity that are observed.

From an interpretability perspective, our approach identifies patterns consistent with semantic competition and collapse but does not provide direct causal explanations of how the model resolves

ambiguity. Similarity-based probes reveal correlations in representation space but cannot determine whether competing interpretations are explicitly represented as distinct alternatives or are entangled in a more complex distributed form.

Ethically, understanding how language models transition from internal uncertainty to confident outputs is important for diagnosing and mitigating hallucinations and misleading responses in deployed systems. At the same time, improved tools for analyzing internal confidence could potentially be misused to design models that appear more persuasive or confident without being more accurate. For this reason, we emphasize that advances in interpretability should be paired with safeguards that promote truthful, uncertainty-aware behavior and responsible deployment.

## Conclusion

- **Similar to abstract but emphasize results.**
- **Summarize main contributions.**

## 9 Conclusion

- **a. Emphasize results**
  - **i.** We show that semantic diversity is high in the mid-layers of GPT-2 XL and then rapidly collapses in later layers, with internal representations converging toward a single dominant meaning.
  - **ii.** This provides direct empirical evidence that transformers resolve ambiguity through an internal competition process rather than making a single immediate commitment.
- **b. Summarize main contributions**
  - **i.** We introduce a geometric framework for measuring semantic diversity and collapse across transformer layers.
  - **ii.** We identify a mid-layer region (layers 17–31) where multiple interpretations co-exist before resolution.
  - **iii.** We connect these internal dynamics to model overconfidence and hallucination, providing a mechanistic foundation for future interpretability and reliability work.

We show that semantic diversity in GPT-2 XL is highest in the mid-layers and then rapidly collapses

in later layers, with internal representations converging toward a single dominant meaning. This provides empirical evidence that transformers resolve ambiguity through an internal process of semantic competition rather than by committing immediately to one interpretation. By tracking this transition across layers, we make visible a core mechanism underlying confident generation.

Our contributions are threefold. First, we introduce a geometric framework for measuring semantic diversity and collapse in transformer hidden states. Second, we identify a mid-layer region (layers 17–31) where multiple interpretations coexist before being resolved. Third, we connect these internal dynamics to overconfidence and hallucination, providing a mechanistic foundation for future work on interpretability and reliability in large language models.

## Abstract

This document is a supplement to the general instructions for \*ACL authors. It contains instructions for using the L<sup>A</sup>T<sub>E</sub>X style files for ACL conferences. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. These instructions should be used both for papers submitted for review and for final versions of accepted papers.

## 10 Introduction

These instructions are for authors submitting papers to \*ACL conferences using L<sup>A</sup>T<sub>E</sub>X. They are not self-contained. All authors must follow the general instructions for \*ACL proceedings,<sup>1</sup> and this document contains additional instructions for the L<sup>A</sup>T<sub>E</sub>X style files.

The templates include the L<sup>A</sup>T<sub>E</sub>X source of this document (`acl_latex.tex`), the L<sup>A</sup>T<sub>E</sub>X style file used to format it (`acl.sty`), an ACL bibliography style (`acl_natbib.bst`), an example bibliography (`custom.bib`), and the bibliography for the ACL Anthology (`anthology.bib`).

## 11 Engines

To produce a PDF file, pdfL<sup>A</sup>T<sub>E</sub>X is strongly recommended (over original L<sup>A</sup>T<sub>E</sub>X plus `dvips+ps2pdf` or `dvipdf`). The style file `acl.sty` can also be used with luaL<sup>A</sup>T<sub>E</sub>X and XeL<sup>A</sup>T<sub>E</sub>X, which are especially suitable for text in non-Latin scripts. The file `acl_lualatex.tex` in this repository provides an example of how to use `acl.sty` with either luaL<sup>A</sup>T<sub>E</sub>X or XeL<sup>A</sup>T<sub>E</sub>X.

## 12 Preamble

The first line of the file must be

```
\documentclass[11pt]{article}
```

To load the style file in the review version:

```
\usepackage[review]{acl}
```

For the final version, omit the review option:

```
\usepackage{acl}
```

To use Times Roman, put the following in the preamble:

```
\usepackage{times}
```

<sup>1</sup><http://acl-org.github.io/ACLPUb/formatting.html>

| Command            | Output | Command             | Output |
|--------------------|--------|---------------------|--------|
| <code>{\`a}</code> | ä      | <code>{\c c}</code> | ç      |
| <code>{\^e}</code> | ê      | <code>{\u g}</code> | ğ      |
| <code>{\`i}</code> | ì      | <code>{\l}</code>   | ł      |
| <code>{\.I}</code> | İ      | <code>{\~n}</code>  | ñ      |
| <code>{\o}</code>  | ø      | <code>{\H o}</code> | ő      |
| <code>{\'u}</code> | ú      | <code>{\v r}</code> | ř      |
| <code>{\aa}</code> | å      | <code>{\ss}</code>  | ß      |

Table 1: Example commands for accented characters, to be used in, *e.g.*, BibT<sub>E</sub>X entries.

(Alternatives like `txfonts` or `newtx` are also acceptable.)

Please see the L<sup>A</sup>T<sub>E</sub>X source of this document for comments on other packages that may be useful.

Set the title and author using `\title` and `\author`. Within the author list, format multiple authors using `\and` and `\And` and `\AND`; please see the L<sup>A</sup>T<sub>E</sub>X source for examples.

By default, the box containing the title and author names is set to the minimum of 5 cm. If you need more space, include the following in the preamble:

```
\setlength\titlebox{<dim>}
```

where `<dim>` is replaced with a length. Do not set this length smaller than 5 cm.

## 13 Document Body

### 13.1 Footnotes

Footnotes are inserted with the `\footnote` command.<sup>2</sup>

### 13.2 Tables and figures

See Table 1 for an example of a table and its caption.

**Do not override the default caption sizes.**

As much as possible, fonts in figures should conform to the document fonts. See Figure 1 for an example of a figure and its caption.

Using the `graphicx` package `graphics` files can be included within figure environment at an appropriate point within the text. The `graphicx` package supports various optional arguments to control the appearance of the figure. You must include it explicitly in the L<sup>A</sup>T<sub>E</sub>X preamble (after the `\documentclass` declaration and before `\begin{document}`) using `\usepackage{graphicx}`.

<sup>2</sup>This is a footnote.



Figure 1: A figure with a caption that runs for more than one line. Example image is usually available through the mwe package without even mentioning it in the preamble.

### 13.3 Hyperlinks

Users of older versions of  $\LaTeX$  may encounter the following error during compilation:

```
\pdfendlink ended up in different nest-
ing level than \pdfstartlink.
```

This happens when pdf $\LaTeX$  is used and a citation splits across a page boundary. The best way to fix this is to upgrade  $\LaTeX$  to 2018-12-01 or later.

### 13.4 Citations

Table 2 shows the syntax supported by the style files. We encourage you to use the natbib styles. You can use the command `\citet` (cite in text) to get “author (year)” citations, like this citation to a paper by ?. You can use the command `\citep` (cite in parentheses) to get “(author, year)” citations (?). You can use the command `\citealp` (alternative cite without parentheses) to get “author, year” citations, which is useful for using citations within parentheses (e.g. ?).

A possessive citation can be made with the command `\citeposs`. This is not a standard natbib command, so it is generally not compatible with other style files.

### 13.5 References

The  $\LaTeX$  and Bib $\TeX$  style files provided roughly follow the American Psychological Association format. If your own bib file is named `custom.bib`, then placing the following before any appendices in your  $\LaTeX$  file will generate the references section for you:

```
\bibliography{custom}
```

You can obtain the complete ACL Anthology as a Bib $\TeX$  file from <https://aclweb.org/>

[anthology/anthology.bib.gz](#). To include both the Anthology and your own .bib file, use the following instead of the above.

```
\bibliography{anthology,custom}
```

Please see Section 14 for information on preparing Bib $\TeX$  files.

### 13.6 Equations

An example equation is shown below:

$$A = \pi r^2 \tag{1}$$

Labels for equation numbers, sections, subsections, figures and tables are all defined with the `\label{label}` command and cross references to them are made with the `\ref{label}` command.

This an example cross-reference to Equation 1.

### 13.7 Appendices

Use `\appendix` before any appendix section to switch the section numbering over to letters. See Appendix A for an example.

## 14 Bib $\TeX$ Files

Unicode cannot be used in Bib $\TeX$  entries, and some ways of typing special characters can disrupt Bib $\TeX$ ’s alphabetization. The recommended way of typing special characters is shown in Table 1.

Please ensure that Bib $\TeX$  records contain DOIs or URLs when possible, and for all the ACL materials that you reference. Use the `doi` field for DOIs and the `url` field for URLs. If a Bib $\TeX$  entry has a URL or DOI field, the paper title in the references section will appear as a hyperlink to the paper, using the `hyperref`  $\LaTeX$  package.

### Limitations

This document does not cover the content requirements for ACL or any other specific venue. Check the author instructions for information on maximum page lengths, the required “Limitations” section, and so on.

### Acknowledgments

This document has been adapted by Steven Bethard, Ryan Cotterell and Rui Yan from the instructions for earlier ACL and NAACL proceedings, including those for ACL 2019 by Douwe Kiela and Ivan Vulić, NAACL 2019 by Stephanie Lukin and Alla Roskovskaya, ACL 2018 by Shay Cohen, Kevin

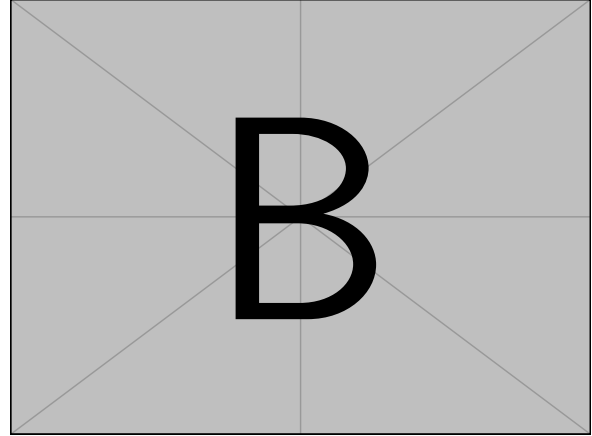
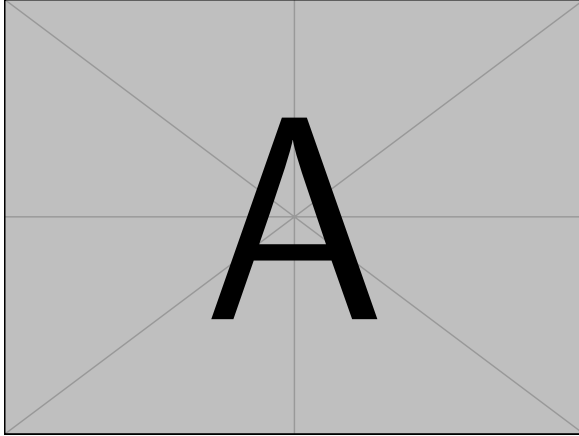


Figure 2: A minimal working example to demonstrate how to place two images side-by-side.

| Output  | natbib command | ACL only command |
|---------|----------------|------------------|
| (?)     | \citep         |                  |
| ?       | \citealp       |                  |
| ?       | \citett        |                  |
| (?)     | \citeyearpar   |                  |
| ?’s (?) |                | \citeposs        |

Table 2: Citation commands supported by the style file. The style is based on the natbib package and supports all natbib citation commands. It also supports commands defined in previous ACL style files for compatibility.

Gimpel, and Wei Lu, NAACL 2018 by Margaret Mitchell and Stephanie Lukin, BibTeX suggestions for (NA)ACL 2017/2018 from Jason Eisner, ACL 2017 by Dan Gildea and Min-Yen Kan, NAACL 2017 by Margaret Mitchell, ACL 2012 by Maggie Li and Michael White, ACL 2010 by Jing-Shin Chang and Philipp Koehn, ACL 2008 by Johanna D. Moore, Simone Teufel, James Allan, and Sadaoki Furui, ACL 2005 by Hwee Tou Ng and Kemal Oflazer, ACL 2002 by Eugene Charniak and Dekang Lin, and earlier ACL and EACL formats written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition*.

## References

- Nura Aljaafari, Danilo S. Carvalho, and Andre Freitas. 2025. Trace: Tracking the emergence of semantic representations in transformers. *arXiv preprint arXiv:2505.17998*.
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## A Example Appendix

### References

### References

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This is an appendix.